

Stochastic Proximal Gradient Consensus Over Time-Varying Multi-Agent Networks

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Presented at INFORMS 2015

Main Content

Setup: Optimization over a time-varying multi-agent network



Main Results

- An algorithm for a large class of convex problems with rate guarantees
- Connections among a number of popular algorithms

Outline

- 1 Review of Distributed Optimization
- 2 The Proposed Algorithm
 - The Proposed Algorithms
 - Distributed Implementation
 - Convergence Analysis
- 3 Connection to Existing Methods
- 4 Numerical Results
- 5 Concluding Remarks

Basic Setup

- Consider the following convex optimization problem

$$\min_{y \in \mathbb{R}} f(y) := \sum_{i=1}^N f_i(y), \quad (\text{P})$$

- Each $f_i(y)$ is a convex and possibly nonsmooth function
- A collection of N agents connected by a network:
 - Network defined by an **undirected** graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$
 - $|\mathcal{V}| = N$ vertices and $|\mathcal{E}| = E$ edges.
 - Each agent can only communicate with its immediate neighbors

Basic Setup

- Numerous applications in optimizing networked systems
 - 1 Cloud computing [Foster *et al* 08]
 - 2 Smart grid optimization [Gan *et al* 13] [Liu-Zhu 14][Kekatos 13]
 - 3 Distributed learning [Mateos *et al* 10] [Boyd *et al* 11] [Bekkerman *et al* 12]
 - 4 Communication and signal processing [Rabbat-Nowak 04] [Schizas *et al* 08] [Giannakis *et al* 15]
 - 5 Seismic Tomography [Zhao *et al* 15]
 - 6 ...

The Algorithms

- A lot of algorithms are available for problem (P)
 - 1 The distributed subgradient (DSG) based methods
 - 2 The Alternating Direction Method of Multiplier (ADMM) based methods
 - 3 The Distributed Dual Averaging based methods
 - 4 ...
- Algorithm families differ in applicable problems and convergence cond.

The DSG Algorithm

- Each agent i keeps a local copy of y , denoted as x_i
- Each agent i iteratively computes

$$x_i^{r+1} = \sum_{j=1}^N w_{ij}^r x_j^r - \gamma^r d_i^r, \forall i \in \mathcal{V}.$$

- We used the following notations
 - 1 $d_i^r \in \partial f_i(y_i^r)$: a subgradient of the local function f_i
 - 2 $w_{ij}^r \geq 0$: the weight for the link e_{ij} at iteration r
 - 3 $\gamma^r > 0$: some stepsize parameter

The DSG Algorithm (Cont.)

- Compactly, the algorithm can be written in vector form

$$x^{r+1} = Wx^r - \gamma^r d^r$$

- 1 $x^r \in \mathbb{R}$: vector of the agents' local variable
- 2 $d^r \in \mathbb{R}$: vector of subgradients
- 3 W : a row-stochastic weight matrix

The DSG Algorithm (Cont.)

- Convergence has been analyzed in many works [Nedić-Ozdaglar 09a][Nedić-Ozdaglar 09b]
- The algorithm converges with a rate of $\mathcal{O}(\ln(r)/\sqrt{r})$ [Chen 12]
- Usually diminishing stepsize
- The algorithm has been generalized to problems with
 - 1 constraints [Nedić-Ozdaglar-Parrilo 10]
 - 2 quantized messages [Nedić *et al* 08]
 - 3 directed graphs [Nedić-Olshevsky 15]
 - 4 stochastic gradients [Ram *et al* 10]
 - 5 ...
- Accelerated versions with rates $\mathcal{O}(\ln(r)/r)$ [Chen 12] [Jakovetić *et al* 14]

The EXTRA Algorithm

- Recently, [Shi *et al* 14] proposed an EXTRA algorithm

$$x^{r+1} = Wx^r - \frac{1}{\beta}d^r + \frac{1}{\beta}d^{r-1} + x^r - \widehat{W}x^{r-1}$$

where $\widehat{W} = \frac{1}{2}(I + W)$; f is assumed to be smooth; W is symmetric

- EXTRA is an error-corrected version of DSG

$$x^{r+1} = Wx^r - \frac{1}{\beta}d^r + \sum_{t=1}^r (W - \widehat{W})x^{t-1}$$

- It is shown that
 - A constant stepsize β can be used (with computable lower bound)
 - The algorithm converges with a (improved) rate of $\mathcal{O}(1/r)$

The ADMM Algorithm

- The general ADMM solves the following two-block optimization problem

$$\begin{aligned} \min_{x,y} \quad & f(x) + g(y) \\ \text{s.t.} \quad & Ax + By = c, \ x \in X, \ y \in Y \end{aligned}$$

- The augmented Lagrangian

$$L(x, y; \lambda) = f(x) + g(y) + \langle \lambda, c - Ax - By \rangle + \frac{\rho}{2} \|c - Ax - By\|^2$$

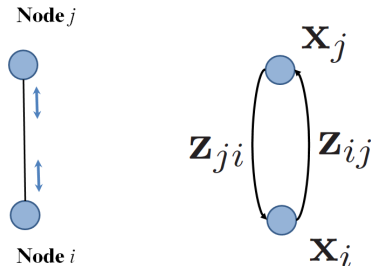
- The algorithm

- 1 Minimize $L(x, y; \lambda)$ w.r.t. x
- 2 Minimize $L(x, y; \lambda)$ w.r.t. y
- 3 $\lambda \leftarrow \lambda + \rho(c - Ax - By)$

The ADMM for Network Consensus

- For each link e_{ij} introduce two **link variables** z_{ij}, z_{ji}
- Reformulate problem (P) as [Schizas *et al* 08]

$$\begin{aligned}
 \min \quad & f(x) := \sum_{i=1}^N f_i(x_i), \\
 \text{s.t.} \quad & x_i = z_{ij}, \quad x_j = z_{ij}, \\
 & x_i = z_{ji}, \quad x_j = z_{ji}, \quad \forall e_{ij} \in \mathcal{E}.
 \end{aligned}$$



The ADMM for Network Consensus (cont.)

- The above problem is equivalent to

$$\begin{aligned} \min \quad & f(x) := \sum_{i=1}^N f_i(x_i), \\ \text{s.t.} \quad & Ax + Bz = 0 \end{aligned} \tag{1}$$

where A, B are matrices related to network topology

- Converges with $\mathcal{O}(1/r)$ rate [Wei-Ozdaglar 13]
- When the objective is smooth and strongly convex, **linear convergence** has been shown in [Shi *et al* 14]
- For a star-network, convergence to stationary solution for nonconvex problem (with rate $\mathcal{O}(1/\sqrt{r})$) [H.-Luo-Razaviyayn 14]

Comparison of ADMM and DSG

Table: Comparison of ADMM and DSG.

	DSG	ADMM
Problem Type	general convex	smooth/smooth+simple NS.
Stepsize	diminishing ^(a)	constant
Convergence Rate	$\mathcal{O}(\ln(r)/\sqrt{r})$	$\mathcal{O}(1/r)$
Network Topology	dynamic	static ^(b)
Subproblem	simple	difficult ^(c)

- (a) Except [Shi *et al* 14], which uses a constant stepsize
- (b) Except [Chang-H.-Wang 14] [Ling *et al* 15], gradient-type subproblem
- (c) Except [Wei-Ozdaglar 13], random graph

Comparison of ADMM and DSG

Connections?



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Setup

- The proposed method is ADMM based
- We consider

$$\min f(y) := \sum_{i=1}^N f_i(y) = \sum_{i=1}^N g_i(y) + h_i(y), \quad (\text{Q})$$

- 1 Each h_i is lower-semicontinuous with **easy “prox” operator**

$$\text{prox}_h^\beta(u) := \min_y h_i(y) + \frac{\beta}{2} \|y - u\|^2.$$

- 2 Each g_i has a Lipschitz continuous gradient, i.e., for some $\rho_i > 0$

$$\|\nabla g_i(y) - \nabla g_i(v)\| \leq \rho_i \|y - v\|, \quad \forall y, v \in \text{dom}(h), \quad \forall i.$$

Graph Structure

- Both **static** and **random time-varying** graph
- For random network assume that
 - 1 At a given iteration $\mathcal{G}^r$ is a subgraph of a connected graph $\mathcal{G}$
 - 2 Each link e has a probability of $p_e \in (0, 1]$ of being active
 - 3 A node i is active if an active link connects to it
 - 4 Each iteration the graph realization is independent

Gradient Information

- Each agent has access to **an estimate** of the gradient $\tilde{g}_i(x_i, \xi_i)$ such that

$$\mathbb{E}[\tilde{g}_i(x_i, \xi_i)] = \nabla g_i(x_i)$$

$$\mathbb{E} \left[\|\tilde{g}_i(x_i, \xi_i) - \nabla g_i(x_i)\|^2 \right] \leq \sigma^2, \quad \forall i$$

- Can be extended to allow only subgradient of the obj

The Augmented Lagrangian

- The problem we solve is still given by

$$\begin{aligned} \min \quad & f(x) := \sum_{i=1}^N g_i(x_i) + h_i(x_i), \\ \text{s.t.} \quad & Ax + Bz = 0 \end{aligned}$$

- The augmented Lagrangian

$$L_{\Gamma}(x, z, \lambda) = \sum_{i=1}^N g_i(x_i) + h_i(x_i) + \langle \lambda, Ax + Bz \rangle + \frac{1}{2} \|Ax + Bz\|_{\Gamma}^2.$$

- A diagonal matrix Γ is used as the penalty parameter (one edge one ρ_{ij})

$$\Gamma := \text{diag}\{\rho_{ij}\}_{ij \in \mathcal{E}}$$

The DySPGC Algorithm

- The proposed algorithm is named DySPGC (Dynamic Stochastic Proximal Gradient Consensus)
- It optimizes $L_{\Gamma}(x, z, \lambda)$ using similar steps as ADMM
- The x -step will be replaced by a proximal gradient step

The DySPGC: Static Graph + Exact Gradient

Algorithm 1. PGC Algorithm

At iteration 0, let $B^T \lambda^0 = 0$, $z^0 = \frac{1}{2} M_+^T x^0$.

At each iteration $r + 1$, update the variable blocks by:

$$x^{r+1} = \arg \min \langle \nabla g(x^r), x - x^r \rangle + h(x) \\ + \frac{1}{2} \left\| Ax + Bz^r + \Gamma^{-1} \lambda^r \right\|_{\Gamma}^2 + \frac{1}{2} \|x - x^r\|_{\Omega}^2$$

$$z^{r+1} = \arg \min \frac{1}{2} \left\| Ax^{r+1} + Bz + \Gamma^{-1} \lambda^r \right\|_{\Gamma}^2$$

$$\lambda^{r+1} = \lambda^r + \Gamma \left(Ax^{r+1} + Bz^{r+1} \right)$$

The DySPGC: Static Graph + Exact Gradient

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 \end{aligned}$$

The DySPGC: Static Graph + Stochastic Gradient

Algorithm 2. SPGC Algorithm

At iteration 0, let $B^T \lambda^0 = 0$, $z^0 = \frac{1}{2} M_+^T x^0$.

At each iteration $r + 1$, update the variable blocks by:

$$\begin{aligned} x^{r+1} &= \arg \min \left\langle \tilde{G}(x^r, \xi^{r+1}), x - x^r \right\rangle + h(x) \\ &\quad + \frac{1}{2} \left\| Ax + Bz^r + \Gamma^{-1} \lambda^r \right\|_{\Gamma}^2 + \frac{1}{2} \|x - x^r\|_{\Omega + \eta^{r+1} I_{MN}}^2 \\ z^{r+1} &= \arg \min \frac{1}{2} \left\| Ax^{r+1} + Bz + \Gamma^{-1} \lambda^r \right\|_{\Gamma}^2 \\ \lambda^{r+1} &= \lambda^r + \Gamma \left(Ax^{r+1} + Bz^{r+1} \right) \end{aligned}$$

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At each iteration $r + 1$, update the variable blocks by:

$$x^{r+1} = \arg \min \left\langle \tilde{G}^{r+1}(x^r, \xi^{r+1}), x - x^r \right\rangle + h^{r+1}(x) \\ + \frac{1}{2} \left\| A^{r+1} x + B^{r+1} z^r + \Gamma^{-1} \lambda^r \right\|_{\Gamma}^2 + \frac{1}{2} \|x - x^r\|_{\Omega^{r+1} + \eta^{r+1} I_{MN}}^2$$

$$x_i^{r+1} = x_i^r, \quad \text{if } i \notin \mathcal{V}^{r+1}$$

$$z^{r+1} = \arg \min \frac{1}{2} \left\| A^{r+1} x^{r+1} + B^{r+1} z + \Gamma^{-1} \lambda^r \right\|_{\Gamma}^2$$

$$z_{ij}^{r+1} = z_{ij}^r, \quad \text{if } e_{ij} \notin \mathcal{A}^{r+1}$$

$$\lambda^{r+1} = \lambda^r + \Gamma \left(A^{r+1} x^{r+1} + B^{r+1} z^{r+1} \right)$$

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Distributed Implementation

- The algorithms admit distributed implementation
- In particular, the PGC admits a **single-variable** characterization

Implementation of PGC

- Define a **stepsize parameter** as

$$\beta_i := \sum_{j \in \mathcal{N}_i} (\rho_{ij} + \rho_{ji}) + \omega_i, \quad \forall i.$$

($\{\omega_i\}$: proximal parameters; $\{\rho_{ij}\}$: penalty parameters for constraints)

- Define a **stepsize matrix** $\mathbf{Y} := \text{diag}([\beta_1, \dots, \beta_N]) \succ 0$.
- Define a **weight matrix** $\mathbf{W} \in \mathbb{R}^{N \times N}$ as (a row-stochastic matrix)

$$(W[i, j]) = \begin{cases} \frac{\rho_{ji} + \rho_{ij}}{\sum_{\ell \in \mathcal{N}_i} (\rho_{\ell i} + \rho_{i\ell}) + \omega_i} = \frac{\rho_{ji} + \rho_{ij}}{\beta_i}, & \text{if } e_{ij} \in \mathcal{E}, \\ \frac{\omega_i}{\sum_{\ell \in \mathcal{N}_i} (\rho_{\ell i} + \rho_{i\ell}) + \omega_i} = \frac{\omega_i}{\beta_i}, & \forall i = j, i \in \mathcal{V} \\ 0, & \text{otherwise,} \end{cases}$$

Implementation of PGC (cont.)

Implementation of PGC

Let $\zeta^r \in \partial h(x^r)$ be some subgradient vector for the nonsmooth function; then the PGC algorithm admits the following single-variable characterization

$$\begin{aligned} x^{r+1} - x^r + \mathbf{Y}^{-1}(\zeta^{r+1} - \zeta^r) \\ = \mathbf{Y}^{-1} \left(-\nabla g(x^r) + \nabla g(x^{r-1}) \right) + Wx^r - \frac{1}{2}(I_N + W)x^{r-1}. \end{aligned}$$

In particular, for smooth problems

$$x^{r+1} = Wx^r - \mathbf{Y}^{-1}\nabla g(x^r) + \mathbf{Y}^{-1}\nabla g(x^{r-1}) + x^r - \frac{1}{2}(I_N + W)x^{r-1}.$$

Convergence Analysis

- We analyze the (rate of) convergence of the proposed methods
- Let us define a matrix of Lip-constants

$$\tilde{P} = \text{diag}([P_1, \dots, P_N]).$$

- Measure convergence rate by [Gao *et al* 14, Ouyang *et al* 14]

$$\underbrace{|f(\bar{x}^r) - f(x^*)|}_{\text{objective gap}}, \quad \text{and} \quad \underbrace{\|A\bar{x}^r + B\bar{z}^r\|}_{\text{consensus gap}}$$

Convergence Analysis

Table: Main Convergence Results.

Algorithm		Convergence Condition	Convergence Rate
Network Type	Gradient Type		
Static	Exact	$YW + Y \succ 2\tilde{P}$	$\mathcal{O}(1/r)$
Static	Stochastic	$YW + Y \succ 2\tilde{P}$	$\mathcal{O}(1/\sqrt{r})$
Random	Exact	$\Omega \succ \tilde{P}$	$\mathcal{O}(1/r)$
Random	Stochastic	$\Omega \succ \tilde{P}$	$\mathcal{O}(1/\sqrt{r})$

Note: For the exact gradient case, stepsize β can be halved if only convergence is needed

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Comparison with Different Algorithms

Algorithm	Conn. with DySPCA	Special Setting
EXTRA [Shi 14]	Special Case	Static, $h \equiv 0$, $W = W^T$, $\tilde{G} = \nabla g$
DSG [Nedić 09]	Different x -step	Static, g smooth, $\tilde{G} = \nabla g$
IC-ADMM [Chang14]	Special Case	Static, $\tilde{G} = \nabla g$, g composite
DLM [Ling 15]	Special Case	Static, $\tilde{G} = \nabla g$, $h \equiv 0$, $\beta_{ij} = \beta$, $\rho_{ij} = \rho$
PG-EXTRA [Shi 15]	Special Case	Static, $W = W^T$, $\tilde{G} = \nabla g$

Comparison with Different Algorithms

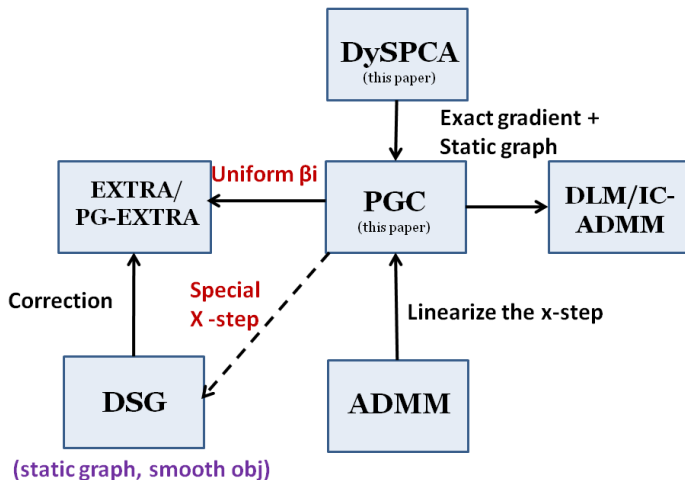


Figure: Relationship among different algorithms

The EXTRA Related Algorithms

- The EXTRA related algorithms (for either smooth or nonsmooth cases) [Shi *et al* 14, 15] are special cases of DySPCA
 - 1 Symmetric weight matrix $W = W^T$
 - 2 Exact gradient
 - 3 Scalar stepsize
 - 4 Static graph

The DSG Method

- Replacing our x -update by (setting the dual variable $\lambda^r = 0$)

$$x^{r+1} = \arg \min \langle \nabla g(x^r), x - x^r \rangle + \langle \mathbf{0}, Ax + Bz^r \rangle \\ + \frac{1}{2} \|Ax + Bz^r\|_{\Gamma}^2 + \frac{1}{2} \|x - x^r\|_{\Omega}^2$$

- Let $\beta_i = \beta_j = \beta$, then the PGC algorithm becomes

$$x^{r+1} = -\frac{1}{\beta} \nabla g(x^r) + \tilde{W}x^r.$$

with $\tilde{W} = \frac{1}{2}(I + W)$

- This is precisely the DSG iterates
- Convergence **not** covered by our results

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Numerical Results

- Some preliminary numerical results by solving a LASSO problem

$$\min_x \frac{1}{2} \sum_{i=1}^N \|A_i x - b_i\|^2 + \nu \|x\|_1$$

where $A_i \in \mathbb{R}^{K \times M}$, $b_i \in \mathbb{R}^K$

- The parameters: $N = 16$, $M = 100$, $\nu = 0.1$, $K = 200$
- Data matrix randomly generated
- Static graphs, generated according to the method proposed in [Yildiz-Scaglione 08], with a radius parameter set to 0.4.

Comparison between PG-EXTRA and PGC

- Stepsize of PG-EXTRA chosen according to conditions given in [Shi 14]
- W is Metropolis constant edge weight matrix
- PCG: $w_i = P_i/2$, $\rho_{ij} = 10^{-3}$

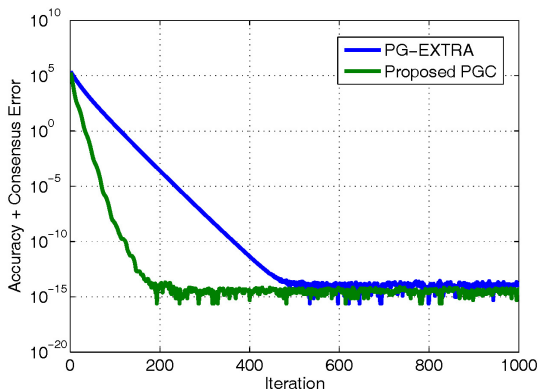


Figure: Comparison between PG-EXTRA and PGC

Comparison between DSG and Stochastic PGC

- Stepsize of DSG chosen as a small constant
- $\sigma^2 = 0.1$
- W is Metropolis constant edge weight matrix
- SPCG: $w_i = P_i, \rho_{ij} = 10^{-3}$

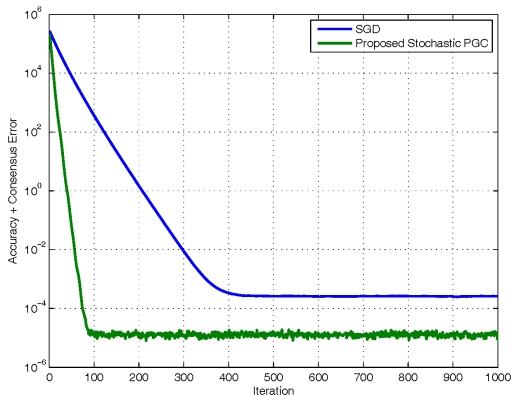


Figure: Comparison between DSG and SPCG

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Summary

- Develop a DySPGC algorithm for multi-agent optimization
- It can deal with
 - 1 Stochastic gradient
 - 2 Time-varying networks
 - 3 Nonsmooth composite objective
- Convergence rate guarantee for various scenarios

Future Work/Generalization

- Identified the relation between DSG-type and ADMM-type methods
- Allows for significant generalization
 - ① Acceleration [Ouyang *et al* 15]
 - ② Variance Reduction for local problem when f_i is a finite sum

$$f_i(x_i) = \sum_{j=1}^M \ell_j(x_i)$$

- ③ Inexact x -subproblems (using, e.g., Conditional-Gradient)
- ④ Nonconvex problems [H.-Luo-Razaviyayn 14]
- ⑤ ...

Thank You!

Parameter Selection

- It is easy to pick various parameters in various different scenarios
- **Case A:** The weight matrix W is given and symmetric
 - 1 We must have $\beta_i = \beta_j = \beta$;
 - 2 For any fixed β , can compute $(\Omega, \{\rho_{ij}\})$
 - 3 Increase β to satisfy convergence condition
- **Case B:** The user has the freedom in picking $(\{\rho_{ij}\}, \Omega)$
 - 1 For any set of $(\{\rho_{ij}\}, \Omega)$, can compute W and $\{\beta_i\}$
 - 2 Increase Ω to satisfy convergence condition
- In either case, the convergence condition can be verified by local agents

Case 1: Exact Gradient with Static Graph

Convergence for PGC Algorithm

Suppose that problem (Q) has a nonempty set of optimal solutions $X^* \neq \emptyset$. Suppose $\mathcal{G}^r = \mathcal{G}$ for all r and $\mathcal{G}$ is connected. Then the PGC converges to a primal-dual optimal solution if

$$2\Omega + M_+ \Xi M_+^T = YW + Y \succ \tilde{P}.$$

- $M_+ \Xi M_+^T$ is some matrix related to network topology
- A sufficient condition is

$$\Omega \succ \tilde{P}$$

or $\omega_i > P_i$ for all $i \in \mathcal{V}$; can be determined locally.

Case 2: Stochastic Gradient with Static Graph

Convergence for SPGC Algorithm

Assume that $\text{dom}(h)$ is a bounded set. Suppose that the following conditions hold

$$\eta^{r+1} = \sqrt{r+1}, \quad \forall r,$$

and the stepsize matrix satisfies

$$2\Omega + M_+ \Xi M_+^T = YW + Y \succ 2\tilde{P}. \quad (8)$$

Then at a given iteration r , we have

$$\mathbb{E} [f(\bar{x}^r) - f(x^*)] + \rho \|A\bar{x}^r + B\bar{z}^r\| \leq \frac{\sigma^2}{\sqrt{r}} + \frac{d_x^2}{2\sqrt{r}} + \frac{1}{2r} \left(d_z^2 + d_\lambda^2(\rho) + \max_i \omega_i d_x^2 \right)$$

where $d_\lambda(\rho) > 0$, $d_x > 0$, $d_z > 0$ are some problem dependent constants.

Case 2: Stochastic Gradient with Static Graph (cont.)

- Both objective value and constraint violation converge with rate $\mathcal{O}(1/\sqrt{r})$
- Easy to extend to the exact gradient case, with rate $\mathcal{O}(1/r)$
- Requires larger proximal parameter Ω than Case 1

Case 3: Exact Gradient with Time-Varying Graph

Convergence for DySPGC Algorithm

Suppose that problem (Q) has a nonempty set of optimal solutions $X^* \neq \emptyset$, and $\tilde{G}(x^r, \xi^{r+1}) = \nabla g(x^r)$ for all r . Suppose the graph is randomly generated. If we choose the following stepsize

$$\Omega \succ \frac{1}{2} \tilde{P}$$

then (x^r, z^r, λ^r) that converges w.p.1. to a primal-dual solution.

- ① The stepsize is more restrictive than Case 1 (not dependent on graph)
- ② Convergence is in the sense of with probability 1

Case 4: Stochastic Gradient with Time-Varying Graph

Convergence for DySPGC Algorithm

Suppose $\{w^t\} = \{x^t, z^t, \lambda^t\}$ is a sequence generated by DySPCA, and that

$$\eta^{r+1} = \sqrt{r+1}, \forall r, \text{ and } \Omega \succ \tilde{P}.$$

Then we have

$$\begin{aligned} & \mathbb{E} [f(\bar{x}^r) - f(x^*) + \rho \|A\bar{x}^r + B\bar{z}^r\|] \\ & \leq \frac{\sigma^2}{\sqrt{r}} + \frac{d_x^2}{2\sqrt{r}} + \frac{1}{2r} \left(2d_J + d_z^2 + d_\lambda^2(\rho) + \max_i \omega_i d_x^2 \right) \end{aligned}$$

where $d_\lambda(\rho), d_J, d_x, d_y$ are some positive constants.